Sparse Polarization Transport and Non-Thin Specialization for Conditional Kuga–Satake Correspondences

Dr. Muralidhara Subbarao
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Abstract

We study the exact finite and specialization steps in *A Conditional Reduction for Algebraic Kuga–Satake Correspondences*. A character-orthogonal selection replaces its 2^r signed square-root transports by fewer than $2r$, with an explicit coefficient-norm bound and no operation on the K3 factor. A separate exact rank factorization compresses a supplied rational two-slot synthesis without assuming that the Clifford algebra splits. We extract a universal exceptional set from the source's proper Hilbert construction. Algebraicity on a nonmeagre set or a set of positive outer measure consequently suffices, and one finite Hilbert format with fixed rational coefficients works throughout the chart. The initial nonzero algebraic correspondence remains a hypothesis. The proofs preserve the full primitive Clifford target, both polarized slots, rational descent and the complete flat class. No new cycle on an unsettled K3 surface is constructed. Contribution assessment: C3/5, linear-size polarization transport and a weaker quantified specialization input with uniform existential cycle format notably sharpen local steps while initial cycle existence remains conditional.

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1 The conditional problem and exact data

Let (S, h) be a primitively amply polarized complex K3 surface, $h^2 = 2d > 0$, and use its actual cup form q. Write

$$V = h^\perp \subset H^2(S, \mathbb{Q}), \qquad T = \mathrm{NS}(S)^\perp_{\mathbb{Q}} \subset V.$$

Then $\dim V = 21$ and its signature is $(2, 19)$. The source [1] assumes, at its generic recovery step,

$$\mathrm{Hg}(V) = \mathrm{SO}(V, q), \qquad \alpha \in \mathrm{CH}^2(S \times B)_{\mathbb{Q}}, \quad 0 \neq \alpha_*|_V : V \longrightarrow H^2(B, \mathbb{Q}), \tag{1}$$

where B is an actual complex abelian variety. The existence of α is not proved by the reduction, by a nonzero Hodge map, by a motivated class, or by this report. Full orthogonal Hodge group implies $T = V$; it is not assumed at the special fibers reached later.

For a rational quadratic Hodge polarization Q of a K3-type structure H, set

$$C(H, Q) = C^+(H, Q) \oplus C^-(H, Q), \qquad x^2 = Q(x, x).$$

Both overall signs of Q are permitted. If the period plane has sign ϵ and an oriented orthogonal normalized basis e_1, e_2, the source uses $J = \epsilon e_1 e_2$ and left multiplication by $\cos\theta + J \sin\theta$. A type (p, q) vector has circle weight $p - q$. The parity spaces $P^\pm_H = C^\pm(H, Q)$ have weight one. The fundamental map is

$$m^Q : H(1) \longrightarrow \mathrm{Hom}(P^+_H, P^-_H), \qquad x \longmapsto L_x. \tag{2}$$

A prescribed invertible odd right multiplier z converts this to $x \mapsto R_z L_x$ on the full even space. A prescribed rational weight-one polarization then converts its dual slot to a second weight-one slot, giving the specified tensor κ. In primitive dimension 21,

$$\dim_{\mathbb{Q}} C^+(V, q) = 2^{20} = 1,048,576.$$

A simple complex spin module has dimension $2^{10} = 1024$ and occurs with multiplicity 1024. These are not interchangeable targets.

Our new contributions concern polarization transport and the quantifiers of specialization. The compression and generator lemmas of the prior report M062 [6] may be used as previously proved finite tools, but are not scored again. The rank-21 Clifford space is not silently normalized to the eight-dimensional rational unit form used there.

2 Linear-size signed-root transport

2.1 The exact source mechanism

Let q_0, q_1 be two rational quadratic Hodge polarizations of the same K3-type structure T, allowing opposite overall signs. Define the rational Hodge endomorphism P by $q_1(u, v) = q_0(Pu, v)$. It is

q_0-self-adjoint. With the sign-adjusted Weil metric, it is self-adjoint for a positive definite real form, hence semisimple with nonzero real eigenvalues. Let

$$P = \sum_{i=1}^{r} \lambda_i p_i, \qquad p_i p_j = \delta_{ij} p_i, \qquad \sum_i p_i = \mathrm{id},$$

over a finite algebraic extension $K/\mathbb{Q}$ containing the distinct eigenvalues and square roots $\nu_i^2 = \lambda_i$. For each sign vector $s \in \{1, -1\}^r$ put

$$R_s = \sum_i s_i \nu_i p_i, \qquad R_s^{-1} = \sum_i s_i \nu_i^{-1} p_i.$$

Then $R_s^2 = P$ and $q_0(R_s u, R_s v) = q_1(u, v)$. The induced parity-preserving Clifford isomorphism $F_s : C(T, q_1)_K \to C(T, q_0)_K$ satisfies

$$A_s(x) := (F_s^-)^{-1} L_x^{q_0} F_s^+ = L_{R_s^{-1} x}^{q_1}. \tag{3}$$

The source's connected joint-Hodge-group argument proves that $F_s^{\pm}$ and their inverses are in the K-span of rational weight-one Hodge maps. In detail, the two induced spin representations cover the same orthogonal action, since R_s commutes with the rational joint Hodge group. Their quotient takes values in the central kernel μ_2 and is the identity on a connected group. Thus their lifts agree. Equivariant Hom spaces are kernels of rational linear equations and commute with scalar extension. This argument, including the spin covering and polarizability, is the source input used below. It does not assert that R_s or p_i acts algebraically on S.

If m^{q_0} is algebraically realized, each A_s is therefore in the K-span of maps realized by rational cycles, using operations on the two abelian factors only. The source recovers the identity through all 2^r signs: $p_i = 2^{-r} \nu_i \sum_s s_i R_s^{-1}$. We need only orthogonality of the selected sign rows.

Theorem 2.1 (Sparse sign reconstruction)**.** *Let $r \geq 1$, let $m = 2^{\lceil \log_2 r \rceil}$, with $m = 1$ when $r = 1$, and choose distinct characters $\chi_1, \ldots, \chi_r$ of $G = (\mathbb{Z}/2\mathbb{Z})^{\log_2 m}$. Set $s(g) = (\chi_1(g), \ldots, \chi_r(g))$ and*

$$c_g = \frac{1}{m} \sum_{i=1}^{r} \nu_i \chi_i(g). \tag{4}$$

Then

$$\sum_{g \in G} c_g R_{s(g)}^{-1} = \mathrm{id}_T, \qquad \sum_{g \in G} c_g A_{s(g)}(x) = L_x^{q_1}. \tag{5}$$

There are $r \leq m < 2r$ selected transports. For each complex embedding of K,

$$\sum_g |c_g|^2 = \frac{1}{m} \sum_i |\nu_i|^2, \qquad \sum_g |c_g| \leq \Big(\sum_i |\nu_i|^2 \Big)^{1/2}. \tag{6}$$

Consequently the algebraicity implication from m^{q_0} to m^{q_1} needs only this selected family before rational descent.

Proof. Distinct real characters are orthogonal: $\sum_g \chi_i(g)\chi_j(g) = m\delta_{ij}$. For the coefficient of p_j in the first sum of (5), substitution gives

$$\frac{1}{m\nu_j} \sum_i \nu_i \sum_g \chi_i(g)\chi_j(g) = 1.$$

Linearity of $x \mapsto L_x^{q_1}$ and (3) give the second identity. Expanding $\sum_g |c_g|^2$ and using the same orthogonality proves its formula, also for nonreal square roots; Cauchy–Schwarz proves the second bound.

Expand each selected $F_s^{\pm}$ and inverse in rational Hodge maps. The two-slot operation is algebraic because such maps come from abelian homomorphisms up to isogeny. Thus the second sum lies in $\mathcal{A} \otimes_{\mathbb{Q}} K$, where $\mathcal{A}$ is the rational space of restricted actions of actual codimension-two cycles. Its value is the rational map m^{q_1}. Choose a rational basis of a complementary space to $\mathcal{A}$; extension of that basis to K shows $\mathcal{A}_K \cap E_{\mathbb{Q}} = \mathcal{A}$ in the full rational comparison space $E_{\mathbb{Q}}$. Equivalently solve the finite rational cycle-column system over $\mathbb{Q}$. This proves algebraic realization without making an algebraic operation on the K3 factor. □

The characters here are a classical Walsh construction, not automorphisms of the K3 surface and not algebraic projectors on individual eigenlines. Selecting ordered eigenvalues need not be Galois invariant. Rational descent is still required. The saving is in the number of algebraic-extension Clifford transports, not a bound of m on rational cycles after expanding the Hodge maps or clearing isogeny denominators. For a formal $r = 21$ spectral list the count is 32 rather than 2^{21}; actual geometric endomorphism constraints may force smaller r. This is an operation-count comparison, not a benchmark on a supplied K3 correspondence.

Proposition 2.2 (Error budget and a count lower bound)**.** *Fix compatible norms on the finite map spaces and a complex embedding. If* $\left\|\widehat{A}_g - A_{s(g)}\right\| \le \delta_g$ *and* $|\widehat{c}_g - c_g| \le \eta_g$*, then*

$$\left\|\sum_g \widehat{c}_g \widehat{A}_g - m^{q_1}\right\| \le \sum_g |c_g|\delta_g + \sum_g \eta_g \left\|\widehat{A}_g\right\|. \tag{7}$$

In particular exact coefficients and a common error δ *give the bound* $(\sum_i |\nu_i|^2)^{1/2}\delta$*. A fixed sign selection which reconstructs* id *for every nonzero spectral-root vector in an open set needs at least* r *transports.*

Proof. Subtract the exact sum, write each difference as $c_g(\widehat{A}_g - A_g) + (\widehat{c}_g - c_g)\widehat{A}_g$, and apply the triangle inequality and (6). For the lower bound let S be the r by m selected-sign matrix. Reconstruction is equivalent to $Sc = \nu$. A proper linear column space cannot contain an open set of root vectors, so $\operatorname{rank} S = r$ and $m \ge r$. □

This lower bound is for universal fixed-sign reconstruction, not an optimality theorem for all geometric methods. A particular ν may require fewer terms. The error bound does not turn a small residual into zero: a known rational denominator and a full-space injectivity bound are separate requirements. Heights of spectral projectors, eigenvalue separation, Clifford frame changes and inverses are not bounded by (6).

2.2 Small exact examples

For $r = 3$, choose the sign columns $(+,+,+), (+,-,+), (+,+,-), (+,-,-)$. With the algebraic-identity fixture $\nu = (1,2,3)$, the weights are $(3/2, 1/2, 0, -1)$. Direct substitution gives $\sum_g c_g s_i(g)/\nu_i = 1$ for each i, $\sum_g |c_g|^2 = 14/4$ and $\sum_g |c_g| = 3 \le \sqrt{14}$. This is not a claim that three rational eigenspaces occur in a simple transcendental Hodge structure.

Here is a rational, genuinely nonscalar quadratic-form fixture. Let $F = \mathbb{Q}[t]/(t^3 - 6t^2 + 9t - 1)$. The polynomial has one root in each of $(0,1), (2,3), (3,4)$, by sign changes, and is irreducible by the rational-root test. On F^2 set

$$q_0((x,y),(x',y')) = \operatorname{Tr}_{F/\mathbb{Q}}((1-t)(xx' + yy')), \quad q_1 = \operatorname{Tr}_{F/\mathbb{Q}}(t(1-t)(xx' + yy')).$$

Both have signature $(2,4)$: at the three real embeddings $1-t$ has signs $+,-,-$, and t is everywhere positive. Choose the oriented positive two-plane at the first embedding as the period plane. Multiplication by t is then a rational Hodge endomorphism P relating these forms. In the basis $1,t,t^2$ on each factor its matrix and Gram blocks are

$$P_0=\begin{pmatrix}0&0&1\\1&0&-9\\0&1&6\end{pmatrix},\quad G_0=\begin{pmatrix}-3&-12&-39\\-12&-39&-129\\-39&-129&-435\end{pmatrix},\quad G_1=\begin{pmatrix}-12&-39&-129\\-39&-129&-435\\-129&-435&-1488\end{pmatrix}. \tag{8}$$

The identities $P_0^tG_0=G_0P_0=G_1$ and $\det G_0=\det G_1=243$ are exact. The three algebraic roots of the polynomial are the eigenvalues of P, each twice; four signed-root transports suffice. No individual spectral projector is asserted rational. This is a polarized rational K3-type structure with explicit forms, not a verified integral K3 embedding or a constructed surface/cycle. Its geometric realization is not used.

3 Rational compression of an existing synthesis

The source produces rational Hodge bases $f_1,\dots,f_a:U\to W_1$ and $g_1,\dots,g_b:U\to W_2$, where $U=H^1(B,\mathbb{Q})$. From the actual input cycle it obtains $\alpha':V\to U\otimes U$ by addition pullback and the mixed Kunneth projector. The map is nonzero because $u\wedge v\mapsto u\otimes v-v\otimes u$ is injective. For a target β, rational descent solves

$$\beta=\sum_{i,j}C_{ij}(f_i\otimes g_j)\alpha',\qquad C\in\mathbb{Q}^{a\times b}. \tag{9}$$

Proposition 3.1 (Compression of the returned coefficient matrix)**.** *For any exact solution C of (9) with $\rho=\operatorname{rank}_{\mathbb{Q}}C$, the same target is realized by at most ρ independent two-slot products of rational Hodge maps. This conclusion requires no rational splitting of the Clifford algebra. Among outer-product decompositions of this fixed matrix C, ρ is minimal.*

Proof. Choose ρ independent columns X of C. Rational elimination gives $Y\in\mathbb{Q}^{\rho\times b}$ with $C=XY$. Writing $X=(x_1,\dots,x_\rho)$ and the rows of Y as y_ℓ^t, define

$$F_\ell=\sum_i(x_\ell)_if_i,\qquad G_\ell=\sum_j(y_\ell)_jg_j.$$

Bilinearity gives $\beta=\sum_{\ell=1}^{\rho}(F_\ell\otimes G_\ell)\alpha'$. Each map is rational Hodge and therefore realized contravariantly by an abelian homomorphism with rational coefficients. These target operations preserve codimension two. A sum of k outer products has matrix rank at most k, which proves the last assertion. □

Different solutions of (9) can have different ranks. Minimizing rank on that affine solution set is not performed here, and a relation among the realized products can make the minimum number of products for β smaller than $\operatorname{rank}C$. Thus this is an exact returned-solution upper bound, not a global minimum-cycle theorem. Clearing denominators in X,Y may also increase the sizes of the corresponding integral homomorphisms. This refinement is a source-specific use of classical rational rank factorization; its value is the exact reduction in the returned synthesis, not a new rank theorem.

4 A universal exceptional set for specialization

Let D be a connected complex polydisc and $p : \mathcal{X} \to D$ a smooth proper holomorphic family of nonempty n-dimensional varieties, $n \geq 2$, with an embedding $\mathcal{X} \hookrightarrow D \times \mathbb{P}^N$. Mark $R^4 p_* \mathbb{Q}$ on this contractible base. The following geometric input is established in the inspected source, `specialization.tex`, lines 12–132: the family is the analytic pullback of a smooth projective algebraic family $\mathcal{U} \to H$ over a reduced quasi-projective base; and top cycle classes of a flat codimension-two subscheme over a smooth parameter are locally constant in the ambient cohomology local system. We use these results, together with proper Hilbert schemes, resolution and the proper mapping theorem.

To recall the reason multiplicities are legitimate, pull a flat subscheme back along an etale coordinate flag on its smooth base. Each coordinate is a nonzerodivisor on the entire subscheme by flatness. Successive Cartier intersections take its top fundamental cycle to the top cycle of the fiber with the scheme lengths included. Restriction of the resulting total cohomology class proves local constancy. Neither purity of every fiber nor componentwise flatness is silently imposed.

Theorem 4.1 (Universal avoidance and bounded format)**.** *For the fixed embedded family there exists a set*

$$E_{\mathcal{X}} = \bigcup_{k \geq 1} E_k \subset D$$

with each E_k proper closed complex analytic, independent of the chosen flat rational class θ and its witness cycles, such that the following holds. If $\theta \in H^0(D, R^4 p_* \mathbb{Q})$ and θ_t is represented by an actual rational codimension-two cycle for even one $t \notin E_{\mathcal{X}}$, then θ_s is algebraic for every $s \in D$.

Moreover, one finite list of Hilbert polynomials $Q_1, \dots, Q_a$ of degree $n-2$ and rational numbers $b_1, \dots, b_a$, chosen once, suffices for all s: there are subschemes $Z_{j,s} \subset \mathcal{X}_s$ of Hilbert polynomial Q_j such that

$$\theta_s = \mathrm{cl}\Big(\sum_{j=1}^{a} b_j [Z_{j,s}]_{n-2} \Big). \tag{10}$$

If $Q_j(k) = d_j k^{n-2}/(n-2)! + \cdots$ and $Lb_j \in \mathbb{Z}$ for an integer $L > 0$, the coefficient denominator L and the weighted-degree bound $\sum_j |b_j| d_j$ are uniform over D. No effective value or global relative cycle section is asserted.

Proof. For each possible Q of degree $n-2$, the relative Hilbert scheme $\mathrm{Hilb}^Q(\mathcal{U}/H)$ is projective over H and has a flat universal subscheme. Resolve its finitely many reduced irreducible components projectively, obtaining smooth irreducible parameters R. They cover the Hilbert scheme set-theoretically and are proper over H. On the smooth projective pullback ambient family over R, the universal subscheme remains flat. The recalled top-cycle lemma applies.

For the analytic classifying map $\eta : D \to H^{\mathrm{an}}$, form $K_R = D \times_{H^{\mathrm{an}}} R^{\mathrm{an}}$. It is proper over D. Marked cycle classes are locally constant on K_R, including its singular or nonreduced parts, since they were constructed before base change. There are countably many Q and finitely many R for each. Each second-countable analytic space K_R has at most countably many closed irreducible components, and each is proper over D. Its image is closed analytic. Define $E_{\mathcal{X}}$ as the union of the images of all such components whose image is not all of D. This construction depends only on the embedded family.

Represent θ_t by a finite rational sum of integral codimension-two subvarieties, $\sum_{j=1}^{a} b_j [Y_{j,t}]$. Each Hilbert point lifts to a point of some K_R. Choose an irreducible component through that point. Because $t \notin E_{\mathcal{X}}$, its image is all of D. The marked class is constant on this connected component. For every $s \in D$, choose any point of it above s. Its universal subscheme is $Z_{j,s}$, with the same

Hilbert polynomial as $Y_{j,t}$ and the same marked top cycle class. GAGA identifies the embedded fiber subschemes with algebraic subschemes of $\mathcal{X}_s$. Keeping the coefficients b_j proves (10) simultaneously for all s.

The leading coefficient of the Hilbert polynomial is the degree of the top cycle, including multiplicities, so it is the same positive integer d_j on every fiber. The coefficient denominator is unchanged. After expanding and collecting integral components, cancellation can only decrease the sum of absolute coefficient times degree. This proves the stated bound. The choices of points over s need not form holomorphic sections, and the avoidance argument gives no algorithm for finding the data. □

Corollary 4.2 (Dichotomy and weaker sufficient hypotheses)**.** *For a fixed flat rational θ, either every fiber class is algebraic, or its algebraicity locus is contained in $E_{\mathcal{X}}$, hence is meagre and has Lebesgue outer measure zero in the full real $2\dim_{\mathbb{C}} D$-dimensional chart. Thus either of the following assumptions suffices: algebraicity on a nonmeagre subset of D, or algebraicity on a subset of positive Lebesgue outer measure. Measurability of that subset is not required.*

Proof. If the algebraicity locus contains a point outside $E_{\mathcal{X}}$, Theorem 4.1 applies. Otherwise it is contained in that set. A proper closed analytic subset of a connected complex manifold has empty interior and measure zero: locally it is contained in the zero set of a nonzero holomorphic function; such a set has real measure zero, for example by local complex-analytic dimension stratification. A countable chart cover and countable subadditivity give outer measure zero. Closed empty-interior sets are nowhere dense. No set contained in their union is nonmeagre or has positive outer measure. □

The proper-image argument and the measure/Baire facts are classical. The weaker quantifiers and the fixed-format consequence are extractions from this particular source proof, not a new unconditional algebraicity principle. These deductions are developed here with the same geometric machinery and with initial cycle existence still conditional.

Corollary 4.3 (One-dimensional limit of the criterion)**.** *For a family as above over a disc, algebraicity of a fixed flat class at uncountably many points suffices for all fibers. In higher-dimensional charts, uncountability alone does not satisfy the avoidance requirement.*

Proof. A proper analytic subset of a disc is discrete and countable, so $E_{\mathcal{X}}$ is countable. In higher dimension a proper divisor can already be uncountable and can contain the entire assumed set. □

A dense countable set can itself be a union of proper analytic point sets and may lie entirely in the exceptional set. Density, including density of a countable special-point locus, is therefore insufficient for this inference. The curve conclusion does not promote a curve family to a full 19-dimensional polarized K3 component.

4.1 Application to the actual K3 input

On a full polarized period polydisc, the source constructs the marked family $\mathcal{X} = \mathcal{S} \times_D \mathcal{A} \times_D \mathcal{A}$, with fixed primitive Clifford data. The full-Hodge-group exception E_{Hg} is also a countable union of proper closed analytic subsets. Suppose that the initial correspondence in (1), without the Hodge-group condition, or the source's surface-cover input exists on a set $A \subset D$ not contained in any countable union of proper analytic subsets. It is enough, in particular, that A be nonmeagre or have positive outer measure. Select $t \in A \setminus (E_{\mathcal{X}} \cup E_{\mathrm{Hg}})$. At this point the source's generic recovery applies.

It is essential to recover the *whole* flat class, not just a restricted action. If (v_i) is a rational basis of V and (q^{ij}) is the inverse cup matrix, it is

$$\theta = \sum_{i,j} q^{ij} v_i \otimes \kappa(v_j) \ \in H^4(\mathcal{S}_t \times \mathcal{A}_t^2, \mathbb{Q}). \tag{11}$$

Precompose the actual cycle with

$$\pi_V = \Delta_S - o \times S - S \times o - \frac{h \times h}{2d}$$

and postcompose with the mixed $H^1 \otimes H^1$ projector. The source has no odd K3 cohomology; these algebraic operations remove the other degrees and the h direction. Poincare duality now identifies its entire class with (11), which is flat in the fixed marking. Theorem 4.1 specializes that class to every point of D, with its fixed coefficient and degree format. The source's restriction to T, then Theorem 2.1 and the stipulated slot identifications, give the prescribed map for every chosen standard datum at the desired fiber.

If this non-thin-input condition is available in a full chart about every desired point of a polarized component, the conclusion holds throughout that component. No assertion is made that such sets of actual witnesses exist. In particular, testing finitely many fibers is not evidence for the measure or category premise.

5 Exact observations and normalization costs

5.1 Nonvanishing is different from exact equality

For a supplied actual cycle $\alpha \in \mathrm{CH}^2(S \times B)_{\mathbb{Q}}$, one rigorous nonzero observation can certify the nonzero-map part of (1). Choose an exact rational $v \in V$ and an exact homology test $\gamma \in H_2(B, \mathbb{Q})$. If an enclosure for $z = \langle \alpha_* v, \gamma \rangle$ gives $|z - \widehat{z}| \le e < |\widehat{z}|$, then $z \neq 0$, hence $\alpha_*|_V \neq 0$. No full frame is required for this one-sided implication. However, the test does not prove that α is algebraic, that the homology identification is exact, or that $\mathrm{Hg}(V) = \mathrm{SO}(V, q)$. At a Picard jump, testing a newly algebraic primitive direction alone need not detect the transcendental part or imply the surface-wedge condition.

Equality with the entire specified tensor is different. With exact actual-cycle columns C and rational target β, one must prove $\beta = Cx$ in the whole relevant rational comparison space. A noninjective moment bank can miss a nonzero difference. If that difference is known to lie in $H^{-1}\mathbb{Z}^b$ and a test map has $\sigma_{\min} \ge s > 0$ on the full space, an error enclosure strictly below s/H proves equality. Without the independently known denominator, arbitrarily small nonzero rationals exist. This is the inspected private original lattice-gap argument [4], not a proof of existence of the initial cycle.

5.2 Moment coordinates and rational synthesis

The private original [5] defines finite moments $m = Ec$, not an unlimited collection of automatically unisolvent tests. For a full-column-rank bank, a specified left inverse reconstructs compatible data; off-range noise requires a metric and an error bound. An infinite-dimensional space has no positive lower frame bound from a fixed finite bank. The actual Hodge/cohomology space and geometric membership must therefore be fixed first.

If the rational Hodge bases in (9) change by invertible matrices S, T, the coefficient matrix changes to $S^{-1}CT^{-t}$. Its rank, and thus Proposition 3.1's returned-solution count, is invariant. Coordinate

conditioning, entry denominators and norms of the individual synthesized homomorphisms need not be invariant. For a physical positive Gram matrix M, the norm in synthesis coordinates R uses R^*MR, not an arbitrarily assigned identity matrix. These are classical coordinate identities, not additional contribution credit.

For uncertain finite pairing data $BC = P$, suppose $\left\|B - \widehat{B}\right\| \le e_B$, $\left\|P - \widehat{P}\right\| \le e_P$, $\left\|\widehat{B}\widehat{C} - \widehat{P}\right\| \le e_{\mathrm{solve}}$, and $\sigma_{\min}(\widehat{B}) \ge s_B > e_B$. Then

$$\left\|C - \widehat{C}\right\| \le \frac{e_P + e_B \left\|\widehat{C}\right\| + e_{\mathrm{solve}}}{s_B - e_B}. \tag{12}$$

Indeed, expand $B(C - \widehat{C})$ into these three errors and use $\sigma_{\min}(B) \ge s_B - e_B$. Finite-panel approximation and unmeasured tail terms must be added to e_P when applicable. Cycle currents in general require a negative Sobolev space rather than an L^2 norm; the finite matrix calculation presupposes well-defined pairings.

5.3 The two slots remain differently typed

For $f : T(1) \to \mathrm{Hom}(P, M)$ and dualities $j_P : P^\vee(-1) \to P$, changing the Hom map to afb with $a : M \to M'$ and $b : P' \to P$ induces the tensor operation

$$a \otimes \left(j_{P'} b^\vee(-1) j_P^{-1}\right). \tag{13}$$

The second factor is not b merely written in the other position. This formula explains why neither a polarization change nor a Clifford conjugation is implemented by an untyped common projection on both tensor slots. The ordinary flip on $H^1 \otimes H^1$ differs from the cohomological graph swap by a minus sign. All such signs and rational scalars remain in the target.

Rational descent may introduce determinants of nonzero cycle-column minors; spectral projectors contain inverse eigenvalue differences; inverse Clifford multipliers contain their quadratic norms; converting rational Hodge maps into integral abelian homomorphisms requires lattice denominators. None of these costs is bounded solely by the number m of selected signs or by $\operatorname{rank} C$. The Hilbert denominator in Theorem 4.1 is uniform only after one successful representative has been chosen, and is not an effective function of degree $2d$.

6 Complete source map and proof boundaries

All ten assigned TeX modules, including the references and dependency table, were literally read in bounded untruncated displays: 1,810 of 1,810 lines. Their architecture is:

Module	Lines	Role
`paper`	46	Master inputs, abstract and conventions
`introduction`	75	Conditional pointwise and component statements
`cycles`	116	Actual cycle operations and surface-cover equivalence
`clifford`	190	Both parities, circle signs, full target and dual slots
`generic`	377	Full Hodge group, spin width, multiplicity recovery
`specialization`	423	Proper Hilbert parameters and complete flat class
`special_data`	243	Restriction and signed-root quadratic-form change
`conclusion`	31	Assembly, including small ranks and Picard jumps
`dependency-table`	163	External geometric and representation inputs
`references`	146	Bibliography, not proof reading of cited works

This complete source reading is not an independent verification of every cited external theorem or a proof of the conditional existence premise. The following safeguards were checked in the source and retained.

6.1 The two geometric input routes

For a smooth connected projective surface Y and a dominant generically finite morphism $f : Y \to S$, the second input is

$$f_* \operatorname{im}\Big[\bigwedge^2 H^0(Y,\Omega^1_Y) \longrightarrow H^0(Y,\Omega^2_Y)\Big] \neq 0.$$

This is trace on holomorphic two-forms, not the assertion that Y merely has positive irregularity. The Albanese map $a : Y \to D$ pulls back all holomorphic one-forms. If $y = f_* a^* \eta \neq 0$, the pairing $\int_D \eta \cup a_* f^* \overline{y} = \int_S y \cup \overline{y}$ shows that the transpose surface correspondence is nonzero on V. Its output has degree $2b-2$, $b = \dim D$.

The source adjusts that degree by an actual divisor square: for an ample L on D let $E = m^*L - p_1^*L - p_2^*L$ on D^2. Its mixed tensor defines an invertible alternating map $M : H^1(D)^\vee \to H^1(D)(1)$. The Koszul sign gives $E^2 = -2\sum_{i<j}(u_i \wedge u_j) \otimes (v_i \wedge v_j)$. Thus $(E^2)_* = -2\bigwedge^2 M$ after Poincare duality, an isomorphism from degree $2b-2$ with the indicated twist to degree two. Composition gives codimension two on $S \times D$. No inverse Lefschetz correspondence on an arbitrary variety is assumed.

Conversely, full $\mathrm{SO}(V)$ makes a nonzero input injective. Its image of a holomorphic two-form is nonzero. Intersect the actual input cycle with ample classes on B to get a rational surface cycle. Hodge–Riemann gives a nonzero pairing; some integral component retains that pairing. A projective resolution gives Y, and nonzero pullback of the K3 two-form forces the map $Y \to S$ to be dominant and generically finite. This uses injectivity on the primitive Hodge structure, not nonvanishing only on a divisor class.

6.2 Full multiplicity recovery, not a rank substitution

The joint Hodge Lie algebra splits into the absolutely simple orthogonal ideal $\mathfrak{g}$ and a commuting kernel. For a highest weight λ of type B_{10}, the projected circle has width $4\lambda_1$. Weight one allows only width at most two, hence only the trivial representation or the spin representation Δ. On the spin part the commuting circle shift is zero. Rational $\mathfrak{g}$-intertwiners to the Clifford parities are therefore Hodge maps, and their rational linear equation kernels commute with complexification.

Under conjugation, $\Delta \otimes \Delta \simeq \bigoplus_{k \text{ even}} \Lambda^k V_{\mathbb{C}}$. Determinant-one diagonal signs show that the vector copy occurs exactly once, in degree 20; it is absent in the other source summands. Consequently the mixed input is $\gamma \otimes t$ with $t \neq 0$ in a multiplicity tensor space. Independent left and right multiplicity maps send that tensor to a spanning set of the full target. Applying the same map on both sides would not justify this span. The source forms rational Hodge-map bases before descending the finite cycle-column system; it does not descend individual complex spin copies.

Proposition 3.1 compresses a solution of that system. It does not prove the representation-width theorem by a matrix rank test, and it does not show that every embedding at a smaller special Hodge group is generated.

6.3 Special data and analytic families

On a marked 19-dimensional polarized polydisc, local Torelli, extension of the polarization line bundle and proper cohomology/base change give a relatively projective K3 family. Fixed rational Clifford polarizations and a symplectic lattice give a holomorphic abelian family. Here the cohomology convention is explicit: $\mathcal{A}_t = (F^1 W_t)^\vee / \operatorname{image}(\Lambda^\vee)$, so $H^1(\mathcal{A}_t, \mathbb{Z}) = \Lambda$. A theta line bundle supplies relative projectivity after shrinking. The fixed marking avoids a monodromy lifting problem on this chart, not on an arbitrary global base.

At a Picard jump $V = T \perp T^\perp$, both Clifford parities of T include as rational Hodge subspaces. A polarization gives a Hodge retraction on the negative parity, and $pL_x^V i = L_x^T$ for $x \in T$. The canonical mask of M062 is an available alternative, not a new result here. Signed-root transport then changes the quadratic form without asserting an algebraic K3 endomorphism. The spin-cover connectedness argument also covers rank two, including opposite-sign forms. For $q_1 = -q_0$ in rank two, $R = i\,\mathrm{id}$ takes $J_1 = -e_1 e_2$ to $-(ie_1)(ie_2) = J_0$.

The external dependencies remain Hodge decomposition and Hodge–Riemann, polarized Hodge semisimplicity, local Torelli, the spin covering and highest-weight theory, Albanese theory, abelian weight-one equivalence, Hilbert boundedness/projectivity, resolution, Cartier pullback, proper analytic mapping, Stein tools and GAGA. The source's dependency table was read, but these full external proofs were not re-audited. No mathematical source correction has been established by this study.

7 Comparison and assessment

Fresh public-title searches on October 8, 2026 located the assigned public manuscript and the primary literature below. Van Geemen's Proposition 6.3 [2] gives the classical Hodge inclusion and its explicit Clifford formula $y \mapsto vye_1$. The inspected discussion distinguishes the left spin action and its commuting right algebra. It supplies background construction, not the initial algebraic cycle in (1).

Floccari–Fu [3], version 2 of February 10, 2026, explicitly distinguish an unspecified algebraic embedding from the prescribed Kuga–Satake embedding in Conjectures 2.4 and 2.6. Their Corollary 4.7 treats the associated K3 surfaces of projective singular OG6 varieties. That geometric result does not provide a very-general rank-21 witness throughout every polarized K3 component. These precise statements were inspected; their complete proofs were not independently audited. No unverified companion global claim is imported as an established theorem.

The private originals were also read afresh: [4], lines 241–307 and 513–694; [5], lines 121–272 and 455–477. This is 424 of 2,050 lines in two actual original files, not a reading of the entire corpus or of their cited additional reports. The unread complements and before/after hashes are in the literal-read ledger. No confidential title, passage, path or report draft was uploaded to search.

7.1 Advantages, limitations and feasible next work

The sparse sign schedule replaces an exponential list by a linear-size list while retaining exact equality, opposite-sign cases and a stable coefficient budget. Its limitation is that algebraic field arithmetic, large Clifford dimensions and rational Hodge-map descent remain. It is not an exponential speedup of the whole cycle construction. The rational rank factorization similarly reduces a supplied operation list, but need not minimize the cycle degree or coefficient height.

The non-thin specialization criterion genuinely weakens the required input set. The fixed-format conclusion also separates an existential uniform denominator/degree bound on a chart from an effective search bound. Neither statement manufactures witnesses or gives an effective way to avoid the exceptional set. Restricting to a countable sampling grid, a special divisor or a curve does not establish the full-chart hypothesis.

A feasible next finite task is to attach the sparse transport to explicit rational Hodge-map bases and compare coefficient heights after rational descent. Another is to optimize the rank of a returned synthesis within its affine solution space, with honest height and integral lattice costs. The separate geometric task is to construct the nonzero input on a set meeting the avoidance criterion. No progress on that last existence problem is asserted here.

7.2 Independent exact checks

The script `_audit/M066-exact-checks.py` uses integers, rational fractions and Gaussian-rational pairs. It passed the following independently executable checks:

Finite fixture family	Cases	Assertions
Walsh orthogonality, reconstruction and error bounds	189	6,020
Clifford transport, both signs and both parities	10	9,366
Rational coefficient rank compression	144	2,484
Worked field, rational example and failure gates	5	39
Total	348	17,909

Cases are parameter families; assertions count individual exact equalities and inequalities. Walsh fixtures run $1 \le r \le 21$. Clifford fixtures use dimensions two through six, not a million-dimensional dense matrix. Rational synthesis fixtures include arbitrary two-slot matrices, but do not assert these are geometric cycles. The Hilbert and measure/category proofs are mathematical arguments using declared background inputs; they are not claims verified by finite tests.

The assessment is C3 for the combined source-specific improvements in transport count and specialization quantifiers, with the explicit bounded-format consequence. Classical character orthogonality, rank factorization and Baire/measure facts are not claimed as new principles. No historical-priority assertion or unconditional Kuga–Satake/Hodge theorem is made. Applicability L3 is a separate inherited catalogue classification.

References

[1] OpenAI, *A Conditional Reduction for Algebraic Kuga–Satake Correspondences*, September 10, 2026. Assigned pinned build, ten TeX modules, 1,810 lines. Pinned public manuscript. Local source identity and exact module hashes accompany this report; no live-branch identity is inferred.

[2] B. van Geemen, *Kuga–Satake varieties and the Hodge conjecture*, arXiv:math/9903146, 1999; published in *The Arithmetic and Geometry of Algebraic Cycles*, 2000, pp. 51–82. Inspected Proposition 6.3 and adjacent conventions. `https://arxiv.org/abs/math/9903146`.

[3] S. Floccari and L. Fu, *The Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties*, arXiv:2504.13607v2, February 10, 2026. Inspected Conjectures 2.4/2.6 and Corollary 4.7. `https://arxiv.org/abs/2504.13607`. Primary author text: `https://irma.math.unistra.fr/~lfu/articles/HCWeilViaOG6.pdf`.

[4] Confidential original Ram manuscript, `TR-item-011-astra.tex`. Finite cycle-period certificates, exact membership and complete rational equality gates; inspected passages specified in Section 7. Local use only.

[5] Confidential original Ram manuscript, `TR-rev-1-old-tr250.tex`. Physical spaces, finite moment coordinates, compatible complexes and polynomial moment construction; inspected passages specified in Section 7. Local use only.

[6] Muralidhara Subbarao, *Exact Clifford Compression and Tensor Certificates on a K3 Quadratic Locus*, Technical Report TR-Ram-OpenAI-722-M062, October 8, 2026. Prior local report; reused compression principles are not counted as new M066 contributions.